

Can Karakoc

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UC Berkeley Computer Science and Data Science graduate with experience building full-stack web applications, AI agent systems, and backend/ML infrastructure end-to-end. Skilled in Python, JavaScript/TypeScript, React, Next.js, FastAPI, SQL, system design, and applied machine learning, with a track record of shipping reliable, well-tested, user-facing products and data-driven systems from scratch. Seeking Software Engineer and Data Engineer/Scientist roles with end-to-end ownership.

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, R

Frontend: React, Next.js, Tailwind CSS, responsive design, Framer Motion

Backend & Systems: FastAPI, REST APIs, async programming, PostgreSQL, NoSQL, caching, ETL pipelines

Data & ML: Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Embeddings, Vector Search (FAISS), Neural Networks

Testing & Tooling: pytest, Git, Linux, Vercel, Render, CI-friendly workflows, Jupyter

Experience

Software Engineer (Contract) – Cendra | May 2026 – July 2026 | Remote

- Architected an autonomous, multi-stage lead pipeline (Python/FastAPI with an APScheduler worker) that ingested, validated, enriched, scored, and synced records end-to-end, scaling throughput from ~25 to 600 leads/day (~24x) and cutting cost per lead ~95%.
- Built LLM classification (GPT-4o) using a hybrid rules-plus-model approach to reach 99.8% accuracy, plus a 0–100 scoring model and a 12-provider enrichment waterfall with automatic Attio CRM sync.
- Productized the system with a Next.js/React/TypeScript dashboard (SWR) delivering real-time pipeline tracking, funnel analytics, live configuration, and provider and data-health monitoring.
- Shipped a companion email-campaign microservice and deployed both systems autonomously on Fly.io and Vercel over Supabase/PostgreSQL, hardened by a pytest suite covering parsing, validation, and edge cases.

Website Content & Design Manager – UC Berkeley Development Engineering | June 2024 – Aug 2025 | Berkeley, CA, USA

- Managed 20+ production web pages for Berkeley's Master of Development Engineering program, improving accessibility, information discoverability, and stakeholder publishing workflows.
- Designed scalable CMS structures and reusable content components, using analytics and user-behavior signals to reduce manual updates and translate stakeholder requirements into maintainable web updates.

Data Analysis Intern – Velocity Inc. | May 2023 – Aug 2023 | Istanbul, Turkey

- Processed millions of mobile-location records through Python pipelines to drive behavioral segmentation, cohort analysis, and product strategy.
- Engineered 20+ temporal and mobility features from raw event-level data; automated weekly analytics workflows and dashboards used by product and strategy teams.

Projects

Protein Interaction Visualizer & Analysis Platform | Personal Project | [App](#) | [GitHub](#)

- Built a full-stack web app (Next.js 14/React + FastAPI) with an interactive 3D visualization UI, computing 18+ structural metrics via a custom Python analysis engine.
- Designed a privacy-first, stateless backend with client-side IndexedDB caching, enabling self-hosted deployment with no cloud storage or user accounts.
- Implemented custom computational geometry algorithms in NumPy/SciPy, backed by 40+ pytest tests covering parsing, metrics, and edge cases; deployed on Vercel + Render.

Vector Similarity Search Engine | Personal Project | [App](#) | [GitHub](#)

- Engineered a full-stack retrieval system that embeds sequence data into dense vector representations and serves nearest-neighbor matches via FAISS similarity search.
- Designed a modular backend pipeline spanning input validation, embedding generation, persistent caching, vector indexing, and ranked retrieval, with a web UI for interactive querying.
- Implemented cache-hit/miss diagnostics and automated tests to reduce redundant computation and validate retrieval correctness across repeated searches.

Reproducible ML Pipeline for High-Dimensional Data Classification | Personal Project | [GitHub](#)

- Developed an end-to-end, reproducible pipeline (dimensionality reduction, autoencoder-based embeddings, classification) achieving 95% accuracy on a high-dimensional dataset.
- Structured the codebase for reproducibility and extensibility, enabling consistent experimentation across preprocessing, modeling, and evaluation stages.

Education

University of California, Berkeley | Berkeley, CA

B.A. Computer Science; B.A. Data Science | Aug 2021 – May 2025

GPA: 3.8/4.0

Relevant Courses: Designing, Visualizing, and Understanding Deep Neural Networks; Introduction to Machine Learning; Data Engineering; Optimization Models in Engineering; Introduction to Database Systems; Machine Structures; Data Structures.

Leadership

Teaching Assistant – Human-centered Design DeCal | May 2023 – Aug 2023 – Led weekly sessions for 60+ students on design methodology, research synthesis, and prototyping.

Design Consultant – Berkeley Innovation | Jan 2023 – May 2024 – Delivered user-centered design solutions for clients through structured research and iteration.